

Propositional and First-Order Logic

Propositional Logic

Propositional Logic

- **Logical constants:** true, false
- **Propositional symbols:** P, Q, S, ... (**atomic sentences**)
- **Wrapping parentheses:** (...)
- Sentences are combined by **connectives**:

$\wedge$...and [conjunction]

$\vee$...or [disjunction]

$\Rightarrow$...implies [implication / conditional]

$\Leftrightarrow$..is equivalent [biconditional]

$\neg$...not [negation]

- **Literal:** atomic sentence or negated atomic sentence

Examples of PL sentences

- P means “It is hot.”
- Q means “It is humid.”
- R means “It is raining.”
- $(P \wedge Q) \rightarrow R$
“If it is hot and humid, then it is raining”
- $Q \rightarrow P$
“If it is humid, then it is hot”
- A better way:
Hot = “It is hot”
Humid = “It is humid”
Raining = “It is raining”

Propositional logic (PL)

- A simple language useful for showing key ideas and definitions
- User defines a set of propositional symbols, like P and Q.
- User defines the **semantics** of each propositional symbol:
 - P means “It is hot”
 - Q means “It is humid”
 - R means “It is raining”
- A sentence (well formed formula) is defined as follows:
 - A symbol is a sentence
 - If S is a sentence, then $\neg S$ is a sentence
 - If S is a sentence, then (S) is a sentence
 - If S and T are sentences, then $(S \vee T)$, $(S \wedge T)$, $(S \rightarrow T)$, and $(S \leftrightarrow T)$ are sentences
 - A sentence results from a finite number of applications of the above rules

Some terms

- The meaning or **semantics** of a sentence determines its **interpretation**.
- Given the truth values of all symbols in a sentence, it can be “evaluated” to determine its **truth value** (True or False).
- A **model** for a KB is a “possible world” (assignment of truth values to propositional symbols) in which each sentence in the KB is True.

More Terms

- A **valid sentence** or **tautology** is a sentence that is True under all interpretations, no matter what the world is actually like or how the semantics are defined. Example: “It’s raining or it’s not raining.”
- An **inconsistent sentence** or **contradiction** is a sentence that is False under all interpretations. The world is never like what it describes, as in “It’s raining and it’s not raining.”
- **P entails Q**, written $P \models Q$, means that whenever P is True, so is Q. In other words, all models of P are also models of Q.

Truth tables

And

p	q	$p \cdot q$
T	T	T
T	F	F
F	T	F
F	F	F

Or

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

If . . . then

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Not

p	$\sim p$
T	F
F	T

Truth tables II

The five logical connectives:

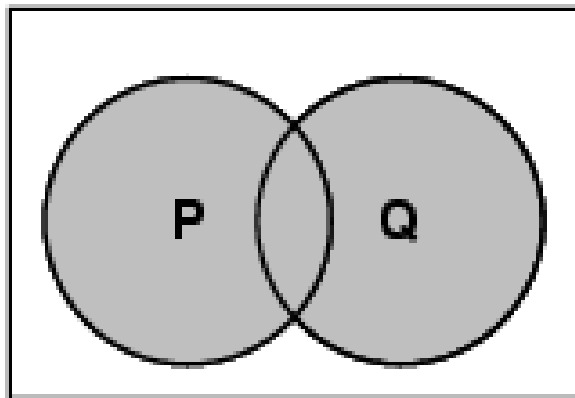
P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>True</i>
<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>
<i>True</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>True</i>	<i>False</i>	<i>False</i>
<i>True</i>	<i>True</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>True</i>	<i>True</i>

A complex sentence:

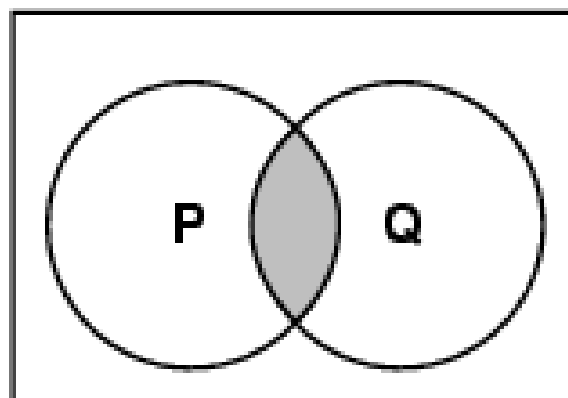
P	H	$P \vee H$	$(P \vee H) \wedge \neg H$	$((P \vee H) \wedge \neg H) \Rightarrow P$
<i>False</i>	<i>False</i>	<i>False</i>	<i>False</i>	<i>True</i>
<i>False</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>True</i>
<i>True</i>	<i>False</i>	<i>True</i>	<i>True</i>	<i>True</i>
<i>True</i>	<i>True</i>	<i>True</i>	<i>False</i>	<i>True</i>

Models of complex sentences

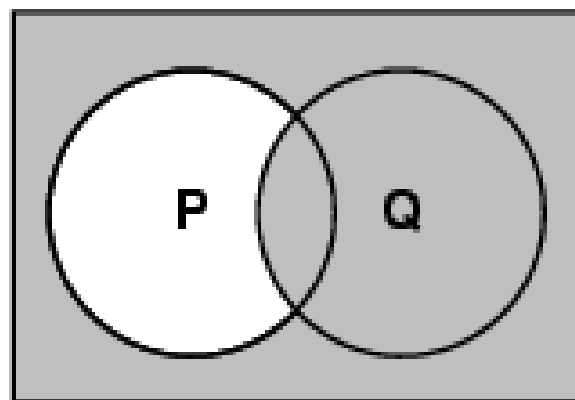
$P \vee Q$



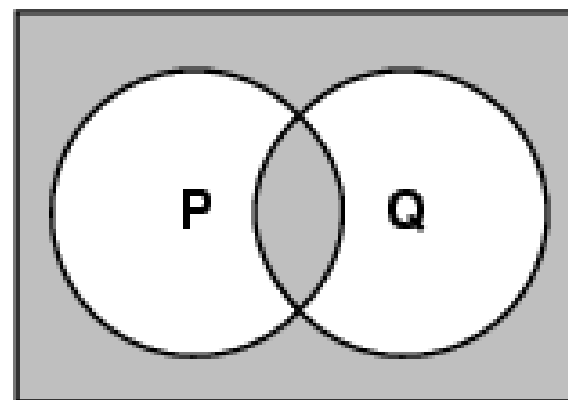
$P \wedge Q$



$P \Rightarrow Q$



$P \Leftrightarrow Q$



Inference rules

- **Logical inference** is used to create new sentences that logically follow from a given set of predicate calculus sentences (KB).
- An inference rule is **sound** if every sentence X produced by an inference rule operating on a KB logically follows from the KB. (That is, the inference rule does not create any contradictions)
- An inference rule is **complete** if it is able to produce every expression that logically follows from (is entailed by) the KB. (Note the analogy to complete search algorithms.)

Sound rules of inference

- Here are some examples of sound rules of inference
 - *A rule is sound if its conclusion is true whenever the premise is true*
- Each can be shown to be sound using a truth table

<u>RULE</u>	<u>PREMISE</u>	<u>CONCLUSION</u>
Modus Ponens	$A, A \rightarrow B$	B
And Introduction	A, B	$A \wedge B$
And Elimination	$A \wedge B$	A
Double Negation	$\neg\neg A$	A
Unit Resolution	$A \vee B, \neg B$	A
Resolution	$A \vee B, \neg B \vee C$	$A \vee C$

Soundness of modus ponens

A	B	$A \rightarrow B$	OK?
True	True	True	✓
True	False	False	✓
False	True	True	✓
False	False	True	✓

Proving things

- A **proof** is a sequence of sentences, where each sentence is either a premise or a sentence derived from earlier sentences in the proof by one of the rules of inference.
- The last sentence is the **theorem** (also called goal or query) that we want to prove.
- Example for the “weather problem” given above.

1 Humid	Premise	“It is humid”
2 Humid $\rightarrow$ Hot	Premise	“If it is humid, it is hot”
3 Hot	Modus Ponens(1,2)	“It is hot”
4 (Hot $\wedge$ Humid) $\rightarrow$ Rain raining”	Premise	“If it’s hot & humid, it’s
5 Hot $\wedge$ Humid	And Introduction(1,3)	“It is hot and humid”
6 Rain	Modus Ponens(4,5)	“It is raining”

Horn sentences

- A **Horn sentence** or **Horn clause** has the form:

$$P_1 \wedge P_2 \wedge P_3 \dots \wedge P_n \rightarrow Q$$

or alternatively

$$\neg P_1 \vee \neg P_2 \vee \neg P_3 \dots \vee \neg P_n \vee Q$$

$$(P \rightarrow Q) = (\neg P \vee Q)$$

where Ps and Q are non-negated atoms

- To get a proof for Horn sentences, apply Modus Ponens repeatedly until nothing can be done

Entailment and derivation

- **Entailment: $KB \models Q$**

- Q is entailed by KB (a set of premises or assumptions) if and only if there is no logically possible world in which Q is false while all the premises in KB are true.
- Or, stated positively, Q is entailed by KB if and only if the conclusion is true in every logically possible world in which all the premises in KB are true.

- **Derivation: $KB \vdash Q$**

- We can derive Q from KB if there is a proof consisting of a sequence of valid inference steps starting from the premises in KB and resulting in Q

Two important properties for inference

Soundness: If $KB \vdash Q$ then $KB \models Q$

- If Q is derived from a set of sentences KB using a given set of rules of inference, then Q is entailed by KB .
- Hence, inference produces only real entailments, or any sentence that follows deductively from the premises is valid.

Completeness: If $KB \models Q$ then $KB \vdash Q$

- If Q is entailed by a set of sentences KB , then Q can be derived from KB using the rules of inference.
- Hence, inference produces all entailments, or all valid sentences can be proved from the premises.

Propositional Logic is a Weak Language

- Hard to identify “individuals” (e.g., Mary, 3)
- Can’t directly talk about properties of individuals or relations between individuals (e.g., “Bill is tall”)
- Generalizations, patterns, regularities can’t easily be represented (e.g., “all triangles have 3 sides”)
- First-Order Logic (abbreviated FOL) is expressive enough to concisely represent this kind of information

FOL adds relations, variables, and quantifiers, e.g.,

- “*Every elephant is gray*”: $\forall x (\text{elephant}(x) \rightarrow \text{gray}(x))$
- “*There is a white alligator*”: $\exists x (\text{alligator}(X) \wedge \text{white}(X))$

Example

- Consider the problem of representing the following information:
 - Every person is mortal.
 - Confucius is a person.
 - Confucius is mortal.
- How can these sentences be represented so that we can infer the third sentence from the first two?

Example II

- In PL we have to create propositional symbols to stand for all or part of each sentence. For example, we might have:
P = “person”; Q = “mortal”; R = “Confucius”
- so the above 3 sentences are represented as:
 $P \rightarrow Q$; $R \rightarrow P$; $R \rightarrow Q$
- Although the third sentence is entailed by the first two, we needed an explicit symbol, R, to represent an individual, Confucius, who is a member of the classes “person” and “mortal”
- To represent other individuals we must introduce separate symbols for each one, with some way to represent the fact that all individuals who are “people” are also “mortal”

Summary

- The process of deriving new sentences from old one is called **inference**.
 - **Sound** inference processes derives true conclusions given true premises
 - **Complete** inference processes derive all true conclusions from a set of premises
- A **valid sentence** is true in all worlds under all interpretations
- If an implication sentence can be shown to be valid, then—given its premise—its consequent can be derived
- Different logics make different **commitments** about what the world is made of and what kind of beliefs we can have regarding the facts
 - Logics are useful for the commitments they do not make because lack of commitment gives the knowledge base engineer more freedom
- **Propositional logic** commits only to the existence of facts that may or may not be the case in the world being represented
 - It has a simple syntax and simple semantics. It suffices to illustrate the process of inference
 - Propositional logic quickly becomes impractical, even for very small worlds

First-Order Logic

Outline

- First-order logic
 - Properties, relations, functions, quantifiers, ...
 - Terms, sentences, axioms, theories, proofs, ...
- Extensions to first-order logic
- Logical agents
 - Reflex agents
 - Representing change: situation calculus, frame problem
 - Preferences on actions
 - Goal-based agents

First-order logic

- First-order Logic (FOL) models the world in terms of
 - **Objects**, which are things with individual identities
 - **Properties** of objects that distinguish them from other objects
 - **Relations** that hold among sets of objects
 - **Functions**, which are a subset of relations where there is only one “value” for any given “input”
- Examples:
 - Objects: Students, lectures, companies, cars ...
 - Relations: Brother-of, bigger-than, outside, part-of, has-color, occurs-after, owns, visits, precedes, ...
 - Properties: blue, oval, even, large, ...
 - Functions: father-of, best-friend, second-half, one-more-than ...

User Provides

- **Constant symbols**, which represent individuals in the world
 - Mary
 - 3
 - Green
- **Function symbols**, which map individuals to individuals
 - father-of(Mary) = John
 - color-of(Sky) = Blue
- **Predicate symbols**, which map individuals to truth values
 - greater(5,3)
 - green(Grass)
 - color(Grass, Green)

FOL Provides

- **Variable symbols**

- E.g., x , y , foo

- **Connectives**

- Same as in PL: not ($\neg$), and ($\wedge$), or ($\vee$), implies ($\rightarrow$), if and only if (biconditional $\leftrightarrow$)

- **Quantifiers**

- Universal $\forall x$ or (Ax)

- Existential $\exists x$ or (Ex)

Sentences are built from terms and atoms

- A **term** (denoting a real-world individual) is a constant symbol, a variable symbol, or an n-place function of n terms.
x and $f(x_1, \dots, x_n)$ are terms, where each x_i is a term.
A term with no variables is a **ground term**
- An **atomic sentence** (which has value true or false) is an n-place predicate of n terms
- A **complex sentence** is formed from atomic sentences connected by the logical connectives:
 $\neg P$, $P \vee Q$, $P \wedge Q$, $P \rightarrow Q$, $P \leftrightarrow Q$ where P and Q are sentences
- A **quantified sentence** adds quantifiers $\forall$ and $\exists$
- A **well-formed formula (wff)** is a sentence containing no “free” variables. That is, all variables are “bound” by universal or existential quantifiers.
 $(\forall x)P(x,y)$ has x bound as a universally quantified variable, but y is free.

Quantifiers

- **Universal quantification**

- $(\forall x)P(x)$ means that P holds for **all** values of x in the domain associated with that variable
- E.g., $(\forall x) \text{dolphin}(x) \rightarrow \text{mammal}(x)$

- **Existential quantification**

- $(\exists x)P(x)$ means that P holds for **some** value of x in the domain associated with that variable
- E.g., $(\exists x) \text{mammal}(x) \wedge \text{lays-eggs}(x)$
- Permits one to make a statement about some object without naming it

Quantifiers

- Universal quantifiers are often used with “implies” to form “rules”:
 $(\forall x) \text{ student}(x) \rightarrow \text{smart}(x)$ means “All students are smart”
- Universal quantification is *rarely* used to make blanket statements about every individual in the world:
 $(\forall x) \text{ student}(x) \wedge \text{smart}(x)$ means “Everyone in the world is a student and is smart”
- Existential quantifiers are usually used with “and” to specify a list of properties about an individual:
 $(\exists x) \text{ student}(x) \wedge \text{smart}(x)$ means “There is a student who is smart”
- A common mistake is to represent this English sentence as the FOL sentence:
 $(\exists x) \text{ student}(x) \rightarrow \text{smart}(x)$
 - But what happens when there is a person who is *not* a student?

Quantifier Scope

- Switching the order of universal quantifiers *does not* change the meaning:
 - $(\forall x)(\forall y)P(x,y) \leftrightarrow (\forall y)(\forall x) P(x,y)$
- Similarly, you can switch the order of existential quantifiers:
 - $(\exists x)(\exists y)P(x,y) \leftrightarrow (\exists y)(\exists x) P(x,y)$
- Switching the order of universals and existentials *does* change meaning:
 - Everyone likes someone: $(\forall x)(\exists y) \text{ likes}(x,y)$
 - Someone is liked by everyone: $(\exists y)(\forall x) \text{ likes}(x,y)$

Connections between All and Exists

We can relate sentences involving $\forall$ and $\exists$ using De Morgan's laws:

$$(\forall x) \neg P(x) \leftrightarrow \neg(\exists x) P(x)$$

$$\neg(\forall x) P \leftrightarrow (\exists x) \neg P(x)$$

$$(\forall x) P(x) \leftrightarrow \neg (\exists x) \neg P(x)$$

$$(\exists x) P(x) \leftrightarrow \neg(\forall x) \neg P(x)$$

Quantified inference rules

- Universal instantiation
 - $\forall x P(x) \therefore P(A)$
- Universal generalization
 - $P(A) \wedge P(B) \dots \therefore \forall x P(x)$
- Existential instantiation
 - $\exists x P(x) \therefore P(F)$ ← **skolem constant F**
- Existential generalization
 - $P(A) \therefore \exists x P(x)$

Universal instantiation (a.k.a. universal elimination)

- If $(\forall x) P(x)$ is true, then $P(C)$ is true, where C is *any* constant in the domain of x
- Example:
 $(\forall x) \text{ eats}(\text{Ziggy}, x) \Rightarrow \text{eats}(\text{Ziggy}, \text{IceCream})$
- The variable symbol can be replaced by any ground term, i.e., any constant symbol or function symbol applied to ground terms only

Existential instantiation

(a.k.a. existential elimination)

- From $(\exists x) P(x)$ infer $P(c)$
- Example:
 - $(\exists x) \text{eats}(\text{Ziggy}, x) \rightarrow \text{eats}(\text{Ziggy}, \text{Stuff})$
- Note that the variable is replaced by a **brand-new constant** not occurring in this or any other sentence in the KB
- Also known as skolemization; constant is a **skolem constant**
- In other words, we don't want to accidentally draw other inferences about it by introducing the constant
- Convenient to use this to reason about the unknown object, rather than constantly manipulating the existential quantifier

Existential generalization (a.k.a. existential introduction)

- If $P(c)$ is true, then $(\exists x) P(x)$ is inferred.
- Example
 $\text{eats}(\text{Ziggy}, \text{IceCream}) \Rightarrow (\exists x) \text{eats}(\text{Ziggy}, x)$
- All instances of the given constant symbol are replaced by the new variable symbol
- Note that the variable symbol cannot already exist anywhere in the expression

Translating English to FOL

Every gardener likes the sun.

$$\forall x \text{ gardener}(x) \rightarrow \text{likes}(x, \text{Sun})$$

You can fool some of the people all of the time.

$$\exists x \forall t \text{ person}(x) \wedge \text{time}(t) \rightarrow \text{can-fool}(x, t)$$

You can fool all of the people some of the time.

$$\forall x \exists t (\text{person}(x) \rightarrow \text{time}(t) \wedge \text{can-fool}(x, t))$$

$$\forall x (\text{person}(x) \rightarrow \exists t (\text{time}(t) \wedge \text{can-fool}(x, t)))$$

← Equivalent

All purple mushrooms are poisonous.

$$\forall x (\text{mushroom}(x) \wedge \text{purple}(x)) \rightarrow \text{poisonous}(x)$$

No purple mushroom is poisonous.

$$\neg \exists x \text{ purple}(x) \wedge \text{mushroom}(x) \wedge \text{poisonous}(x)$$

$$\forall x (\text{mushroom}(x) \wedge \text{purple}(x)) \rightarrow \neg \text{poisonous}(x)$$

← Equivalent

There are exactly two purple mushrooms.

$$\begin{aligned} &\exists x \exists y \text{ mushroom}(x) \wedge \text{purple}(x) \wedge \text{mushroom}(y) \wedge \text{purple}(y) \wedge \neg(x=y) \wedge \forall z \\ &(\text{mushroom}(z) \wedge \text{purple}(z)) \rightarrow ((x=z) \vee (y=z)) \end{aligned}$$

Clinton is not tall.

$$\neg \text{tall}(\text{Clinton})$$

X is above Y iff X is on directly on top of Y or there is a pile of one or more other objects directly on top of one another starting with X and ending with Y.

$$\forall x \forall y \text{ above}(x, y) \leftrightarrow (\text{on}(x, y) \vee \exists z (\text{on}(x, z) \wedge \text{above}(z, y)))$$

Semantics of FOL

- **Domain M :** the set of all objects in the world (of interest)
- **Interpretation I :** includes
 - Assign each constant to an object in M
 - Define each function of n arguments as a mapping $M^n \Rightarrow M$
 - Define each predicate of n arguments as a mapping $M^n \Rightarrow \{T, F\}$
 - Therefore, every ground predicate with any instantiation will have a truth value
 - In general there is an infinite number of interpretations because $|M|$ is infinite
- **Define logical connectives:** $\sim, \wedge, \vee, \Rightarrow, \Leftrightarrow$ as in PL
- **Define semantics of $(\forall x)$ and $(\exists x)$**
 - $(\forall x) P(x)$ is true iff $P(x)$ is true under all interpretations
 - $(\exists x) P(x)$ is true iff $P(x)$ is true under some interpretation

- **Model:** an interpretation of a set of sentences such that every sentence is *True*
- **A sentence is**
 - **satisfiable** if it is true under some interpretation
 - **valid** if it is true under all possible interpretations
 - **inconsistent** if there does not exist any interpretation under which the sentence is true
- **Logical consequence:** $S \models X$ if all models of S are also models of X

Axioms, definitions and theorems

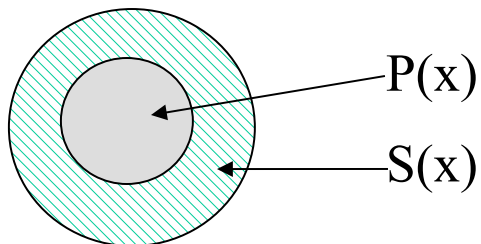
- **Axioms** are facts and rules that attempt to capture all of the (important) facts and concepts about a domain; axioms can be used to prove **theorems**
 - Mathematicians don't want any unnecessary (dependent) axioms –ones that can be derived from other axioms
 - Dependent axioms can make reasoning faster, however
 - Choosing a good set of axioms for a domain is a kind of design problem
- A **definition** of a predicate is of the form " $p(X) \leftrightarrow \dots$ " and can be decomposed into two parts
 - Necessary** description: " $p(x) \rightarrow \dots$ "
 - Sufficient** description " $p(x) \leftarrow \dots$ "
 - Some concepts don't have complete definitions (e.g., $\text{person}(x)$)

More on definitions

- A **necessary** condition must be satisfied for a statement to be true.
- A **sufficient** condition, if satisfied, assures the statement's truth.
- Duality: “P is sufficient for Q” is the same as “Q is necessary for P.”
- Examples: define $\text{father}(x, y)$ by $\text{parent}(x, y)$ and $\text{male}(x)$
 - $\text{parent}(x, y)$ is a necessary (**but not sufficient**) description of $\text{father}(x, y)$
 - $\text{father}(x, y) \rightarrow \text{parent}(x, y)$
 - $\text{parent}(x, y) \wedge \text{male}(x) \wedge \text{age}(x, 35)$ is a **sufficient (but not necessary)** description of $\text{father}(x, y)$:
$$\text{father}(x, y) \leftarrow \text{parent}(x, y) \wedge \text{male}(x) \wedge \text{age}(x, 35)$$
 - $\text{parent}(x, y) \wedge \text{male}(x)$ is a **necessary and sufficient** description of $\text{father}(x, y)$
$$\text{parent}(x, y) \wedge \text{male}(x) \leftrightarrow \text{father}(x, y)$$

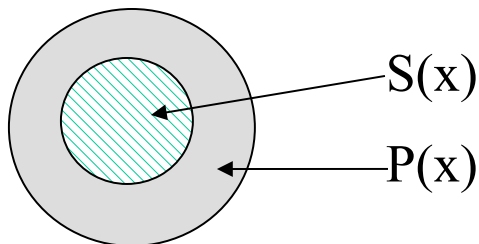
More on definitions

$S(x)$ is a
necessary
condition of $P(x)$



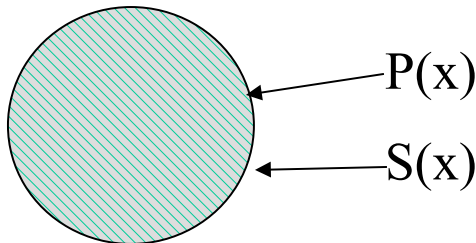
$$(\forall x) P(x) \Rightarrow S(x)$$

$S(x)$ is a
sufficient
condition of $P(x)$



$$(\forall x) P(x) \Leftarrow S(x)$$

$S(x)$ is a
necessary and
sufficient
condition of $P(x)$



$$(\forall x) P(x) \Leftrightarrow S(x)$$

Higher-order logic

- FOL only allows to quantify over variables, and variables can only range over objects.
- HOL allows us to quantify over relations
- Example: (quantify over functions)
“two functions are equal iff they produce the same value for all arguments”
$$\forall f \forall g (f = g) \leftrightarrow (\forall x f(x) = g(x))$$
- Example: (quantify over predicates)
$$\forall r \text{ transitive}(r) \rightarrow (\forall xyz) r(x,y) \wedge r(y,z) \rightarrow r(x,z)$$
- More expressive, but **undecidable**. (there isn't an effective algorithm to decide whether all sentences are valid)
 - First-order logic is decidable only when it uses predicates with only one argument.

Logical agents for the Wumpus World

Three (non-exclusive) agent architectures:

- Reflex agents

- Have rules that classify situations, specifying how to react to each possible situation

- Model-based agents

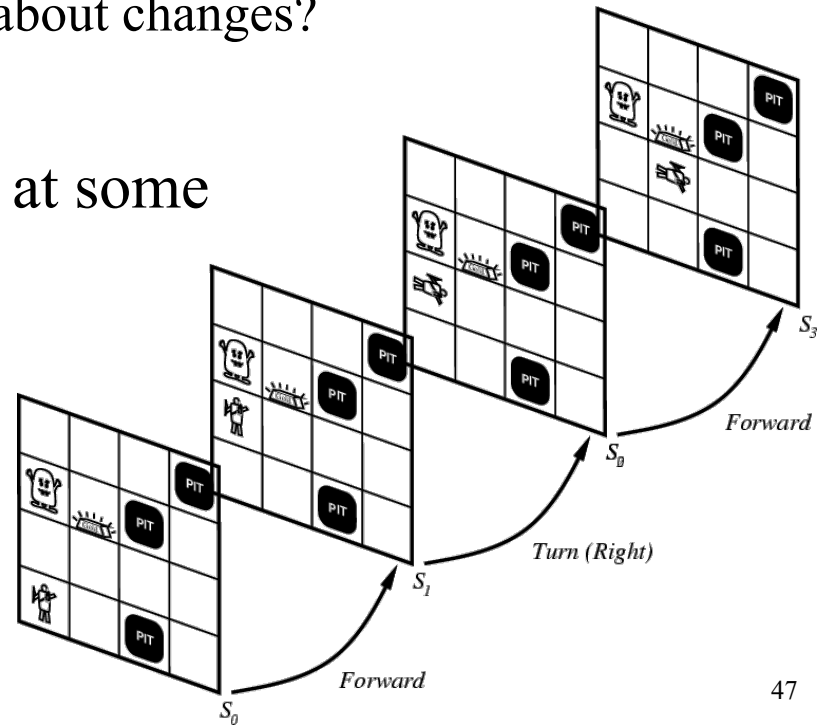
- Construct an internal model of their world

- Goal-based agents

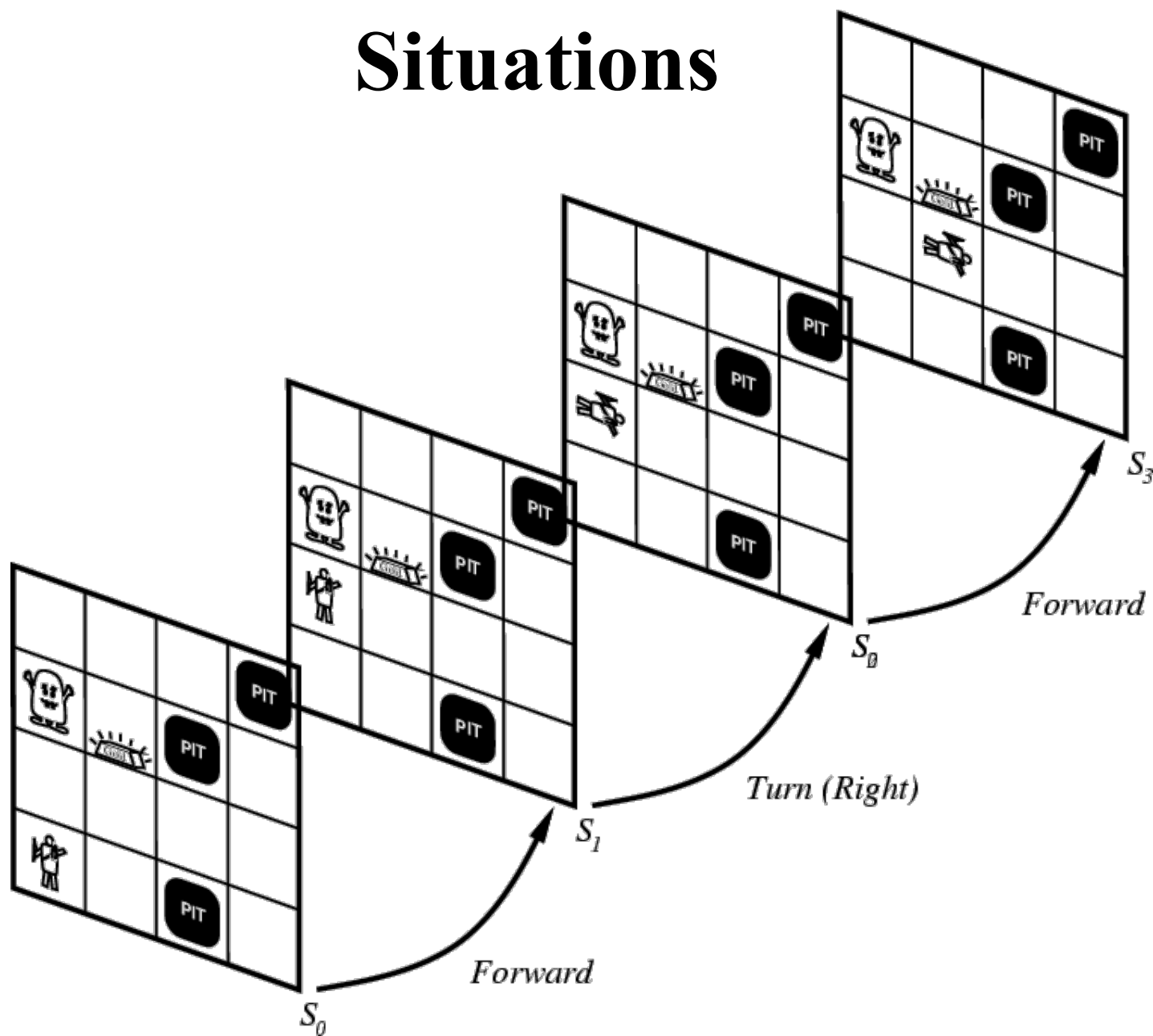
- Form goals and try to achieve them

Representing change

- Representing change in the world in logic can be tricky.
- One way is just to change the KB
 - Add and delete sentences from the KB to reflect changes
 - How do we remember the past, or reason about changes?
- **Situation calculus** is another way
- A **situation** is a snapshot of the world at some instant in time
- When the agent performs an action A in situation S1, the result is a new situation S2.



Situations



Situation calculus

- A **situation** is a snapshot of the world at an interval of time during which nothing changes
- Every true or false statement is made with respect to a particular situation.
 - Add **situation variables** to every predicate.
 - $\text{at}(\text{Agent}, 1, 1)$ becomes $\text{at}(\text{Agent}, 1, 1, s_0)$: $\text{at}(\text{Agent}, 1, 1)$ is true in situation (i.e., state) s_0 .
 - Alternatively, add a special 2nd-order predicate, **holds(f,s)**, that means “f is true in situation s.” E.g., $\text{holds}(\text{at}(\text{Agent}, 1, 1), s_0)$
- Add a new function, **result(a,s)**, that maps a situation s into a new situation as a result of performing action a. For example, $\text{result}(\text{forward}, s)$ is a function that returns the successor state (situation) to s
- Example: The action agent-walks-to-location-y could be represented by
 - $(\forall x)(\forall y)(\forall s) (\text{at}(\text{Agent}, x, s) \wedge \neg \text{onbox}(s)) \rightarrow \text{at}(\text{Agent}, y, \text{result}(\text{walk}(y), s))$

Knowledge engineering!

- Modeling the “right” conditions and the “right” effects at the “right” level of abstraction is very difficult
- Knowledge engineering (creating and maintaining knowledge bases for intelligent reasoning) is an entire field of investigation
- Many researchers hope that automated knowledge acquisition and machine learning tools can fill the gap:
 - Our intelligent systems should be able to **learn** about the conditions and effects, just like we do!
 - Our intelligent systems should be able to learn when to pay attention to, or reason about, certain aspects of processes, depending on the context!

Preferences among actions

- A problem with the Wumpus world knowledge base that we have built so far is that it is difficult to decide which action is best among a number of possibilities.
- For example, to decide between a forward and a grab, axioms describing when it is OK to move to a square would have to mention glitter.
- This is not modular!
- We can solve this problem by **separating facts about actions from facts about goals**. This way our **agent can be reprogrammed just by asking it to achieve different goals**.

Preferences among actions

- The first step is to describe the desirability of actions independent of each other.
- In doing this we will use a simple scale: actions can be Great, Good, Medium, Risky, or Deadly.
- Obviously, the agent should always do the best action it can find:

$$(\forall a,s) \text{ Great}(a,s) \rightarrow \text{Action}(a,s)$$

$$(\forall a,s) \text{ Good}(a,s) \wedge \neg(\exists b) \text{ Great}(b,s) \rightarrow \text{Action}(a,s)$$

$$(\forall a,s) \text{ Medium}(a,s) \wedge (\neg(\exists b) \text{ Great}(b,s) \vee \text{ Good}(b,s)) \rightarrow \text{Action}(a,s)$$

...

Preferences among actions

- We use this action quality scale in the following way.
- Until it finds the gold, the basic strategy for our agent is:
 - Great actions include picking up the gold when found and climbing out of the cave with the gold.
 - Good actions include moving to a square that's OK and hasn't been visited yet.
 - Medium actions include moving to a square that is OK and has already been visited.
 - Risky actions include moving to a square that is not known to be deadly or OK.
 - Deadly actions are moving into a square that is known to have a pit or a Wumpus.

Goal-based agents

- Once the gold is found, it is necessary to change strategies. So now we need a new set of action values.
- We could encode this as a rule:
 - $(\forall s) \text{Holding}(\text{Gold}, s) \rightarrow \text{GoalLocation}([1,1], s)$
- We must now decide how the agent will work out a sequence of actions to accomplish the goal.
- Three possible approaches are:
 - **Inference**: good versus wasteful solutions
 - **Search**: make a problem with operators and set of states
 - **Planning**: to be discussed later